AI Applications for Engineers

Dr. Rami Alloush

March 2026

Use AI *wisely* (3Ds)

Discuss

Talk with your advisor or leader about related policies and compliance rules on using AI.

Disclose

Provide a detailed log of your method and tool usage and even prompt if you can.

Document

Always provide explicit disclosure when AI tools are integrated into your work.

Today’s Session

It is NOT

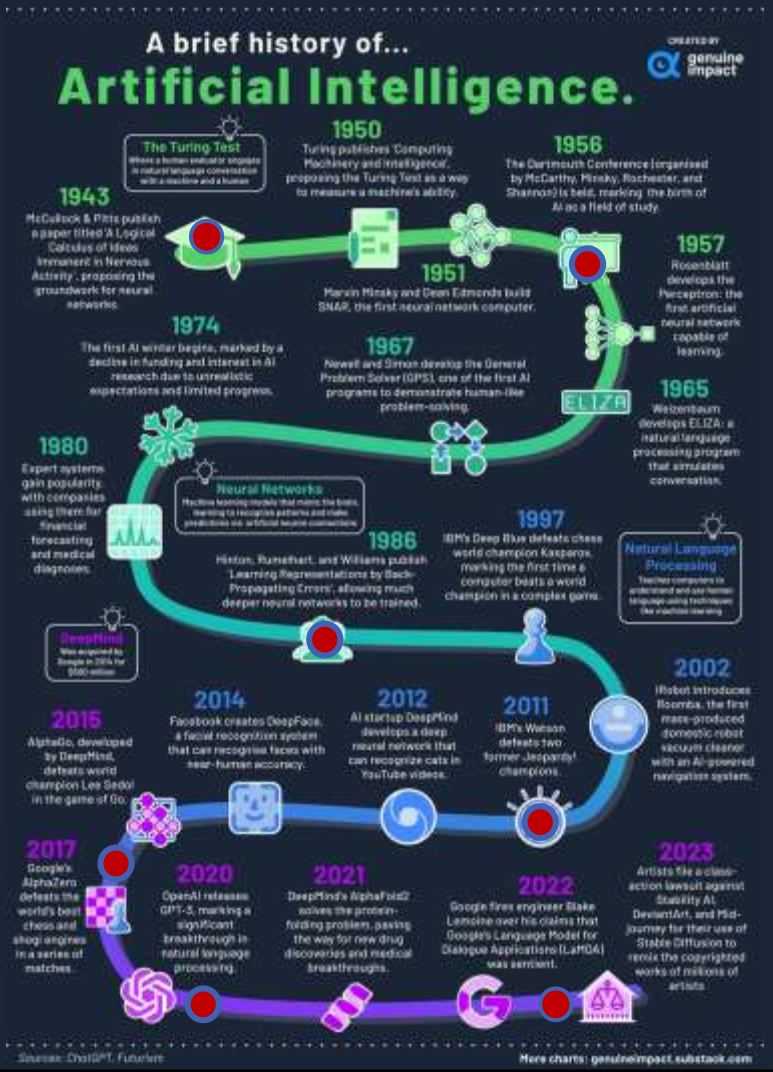
- 🚫 Step-by-step tutorial
- 🚫 Specific workflow in-depth
- 🚫 Direct instructions
- 🚫 Deep science and math of AI

We WILL

- ✅ Brief history of AI
- ✅ Cover core concepts
- ✅ Understand the sudden interest!
- ✅ Sample of practical use cases

Brief history of AI

Key Milestones



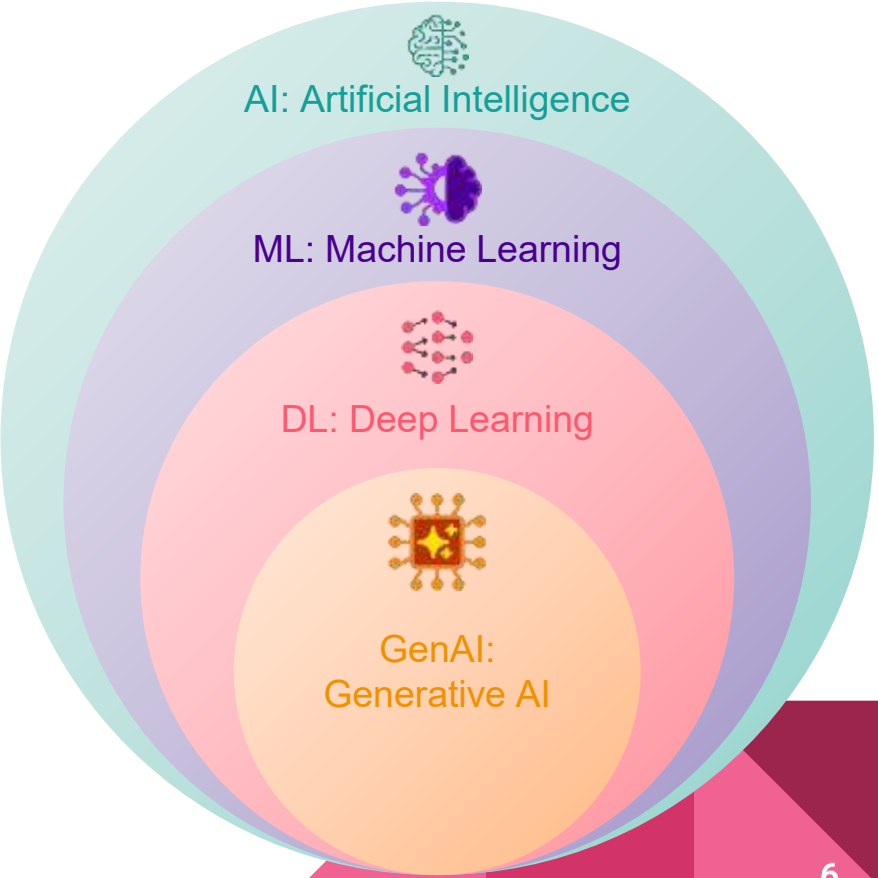
Brief history of AI

Turing Test	AI: Artificial Intelligence	ML: Machine Learning	IBM Deep Blue	DL: Deep Learning	AlexNet	GenAI: Generative AI
1950	1956	1959	1997	2012	2017	2022

** Dates denotes when the term becomes mainstream*

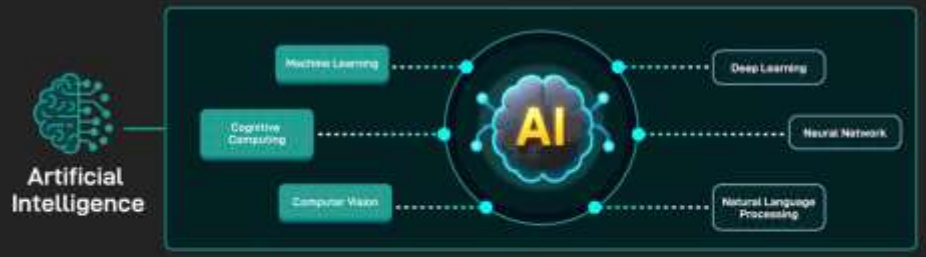
AI > ML > DL > GenAI

- AI - Artificial Intelligence
  - Broad concept of machines doing intelligent tasks
- ML - Machine Learning
  - Subset of AI using algorithms to learn from data
- DL - Deep Learning
  - Subset of ML using deep layered neural networks
- GenAI - Generative AI
  - Subset of DL that creates new content and data



AI - Artificial Intelligence

Broad concept of machines performing intelligent tasks

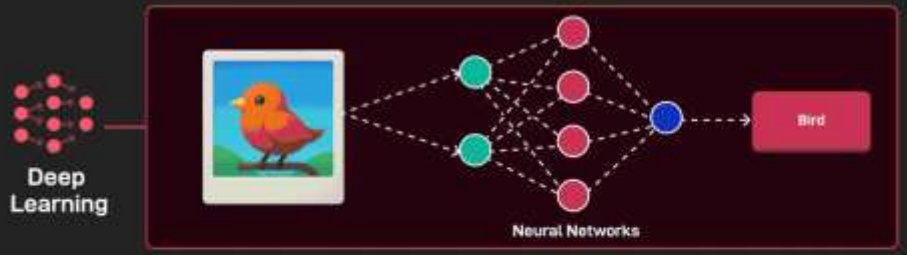


ML - Machine Learning

Subset of AI using algorithms to learn from data

DL - Deep Learning

Subset of ML using neural networks with multiple layers



GenAI - Generative AI

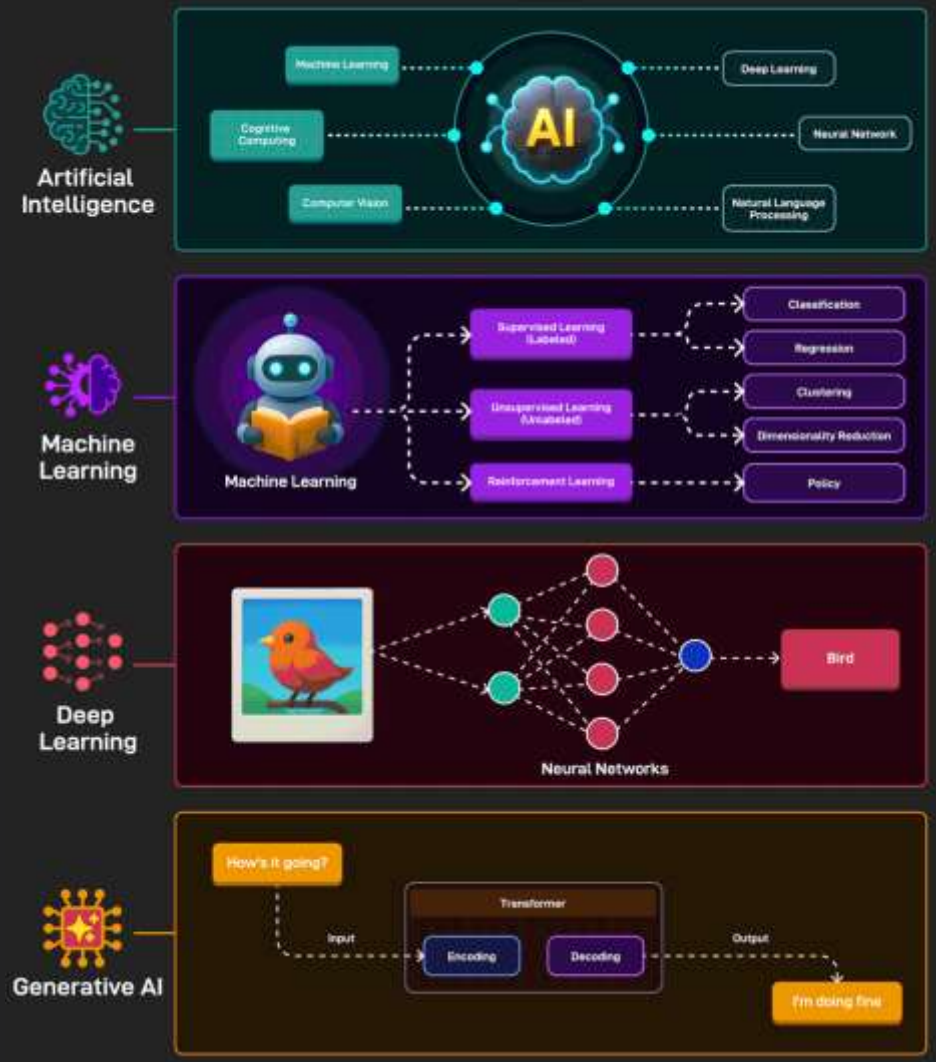
Subset of DL that creates new content and data

https://blog.bytebytego.com/p/ep168-ai-vs-machine-learning-vs-deep

Scenario

You design a system for a server room. It has a lookup table: "If temp > 25°C AND humidity > 60%, increase fan speed by 20%."

- **Answer: Traditional AI** (Expert System)
- **Why?** It mimics a human decision-maker but doesn't "learn."

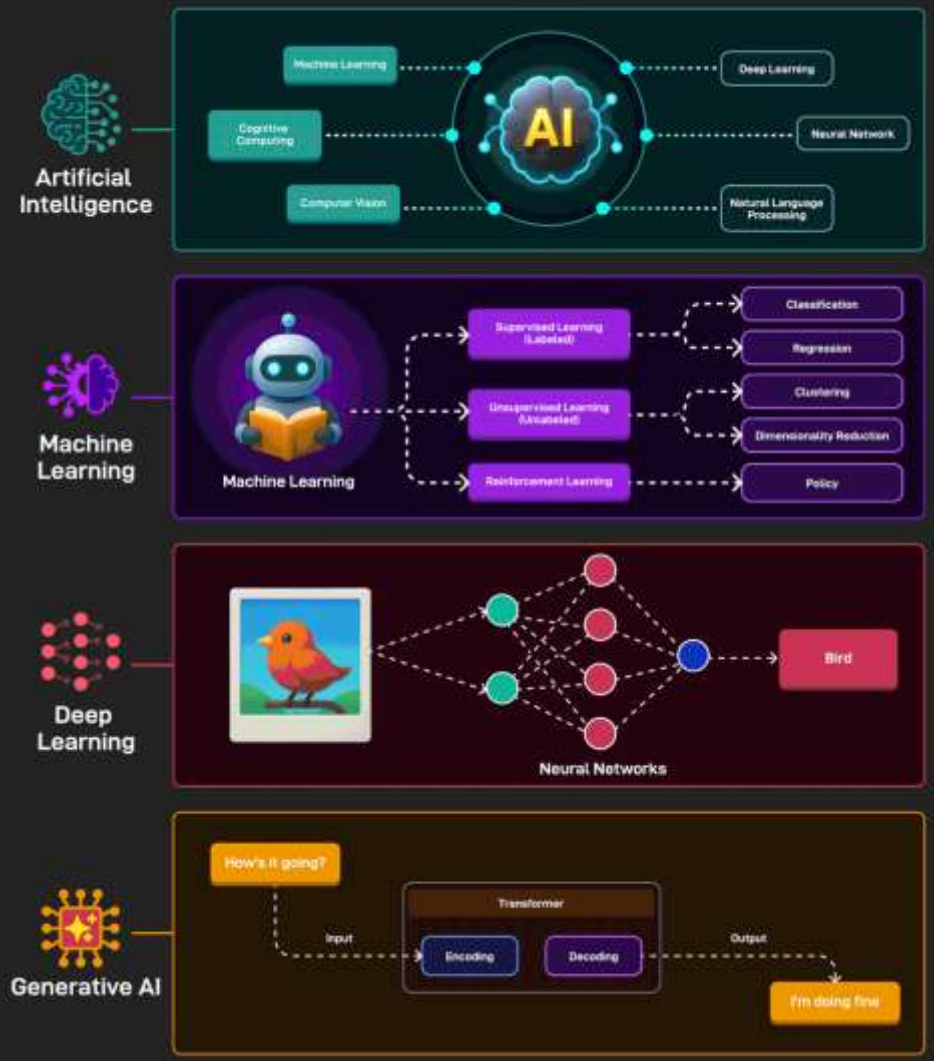


https://blog.bytebytego.com/p/ep168-ai-vs-machine-learning-vs-deep

Scenario

You use a high-speed camera on a production line. A Neural Network analyzes the pictures of every weld to find microscopic fractures that a human eye might miss.

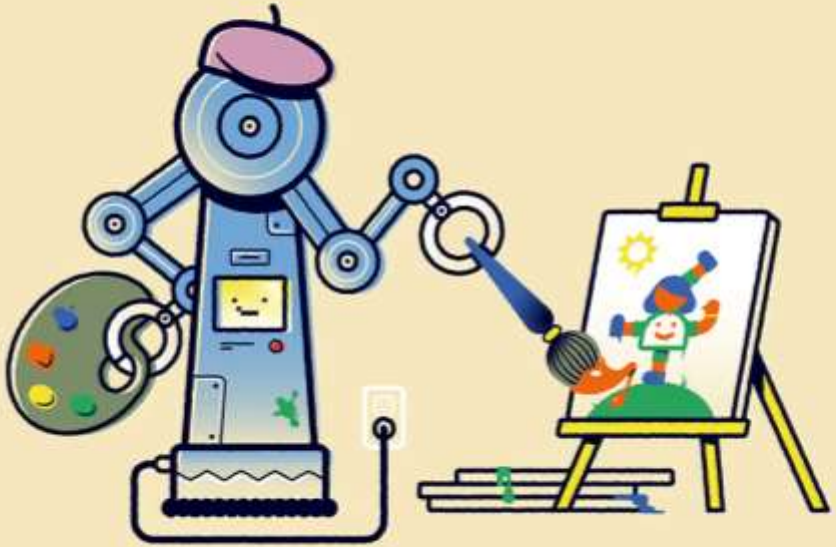
- **The Answer: Deep Learning (DL)**
- **Why?** It processes raw, unstructured pixels. The network figured fractures out.



https://blog.bytebytego.com/p/ep168-ai-vs-machine-learning-vs-deep

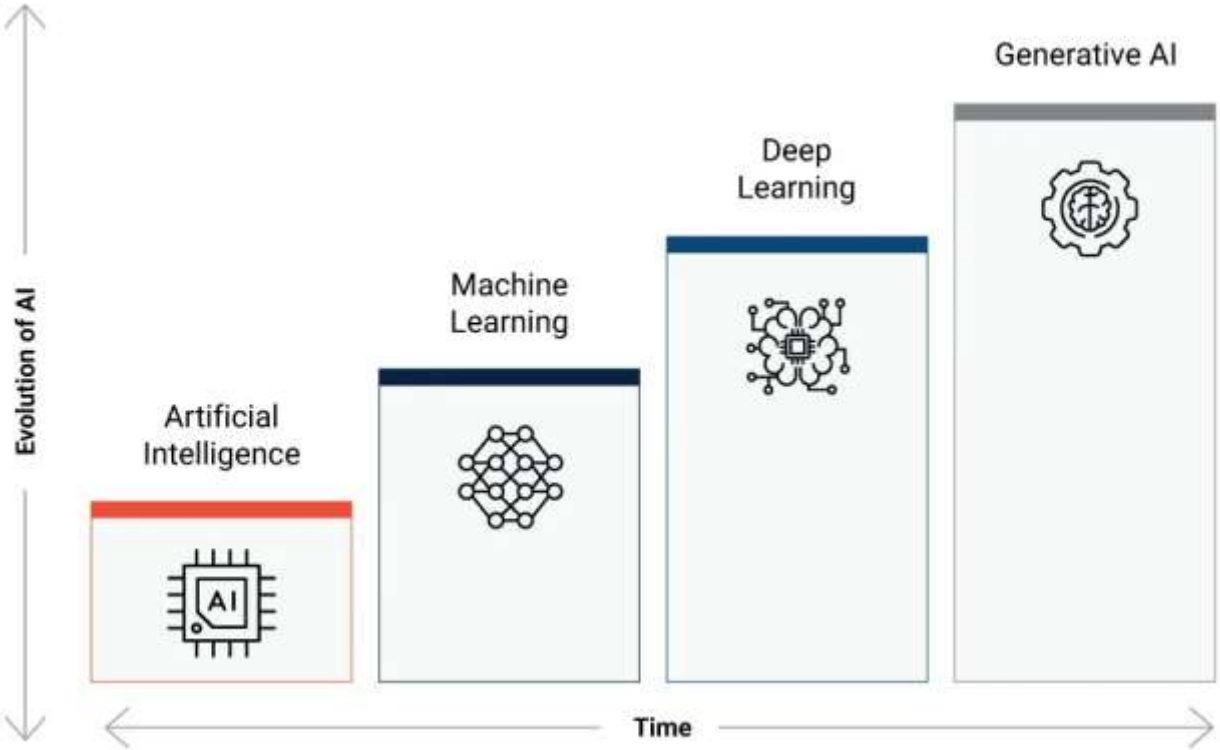
Generative AI

Why is the sudden interest?

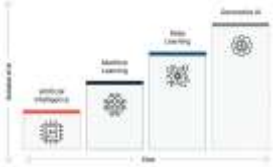


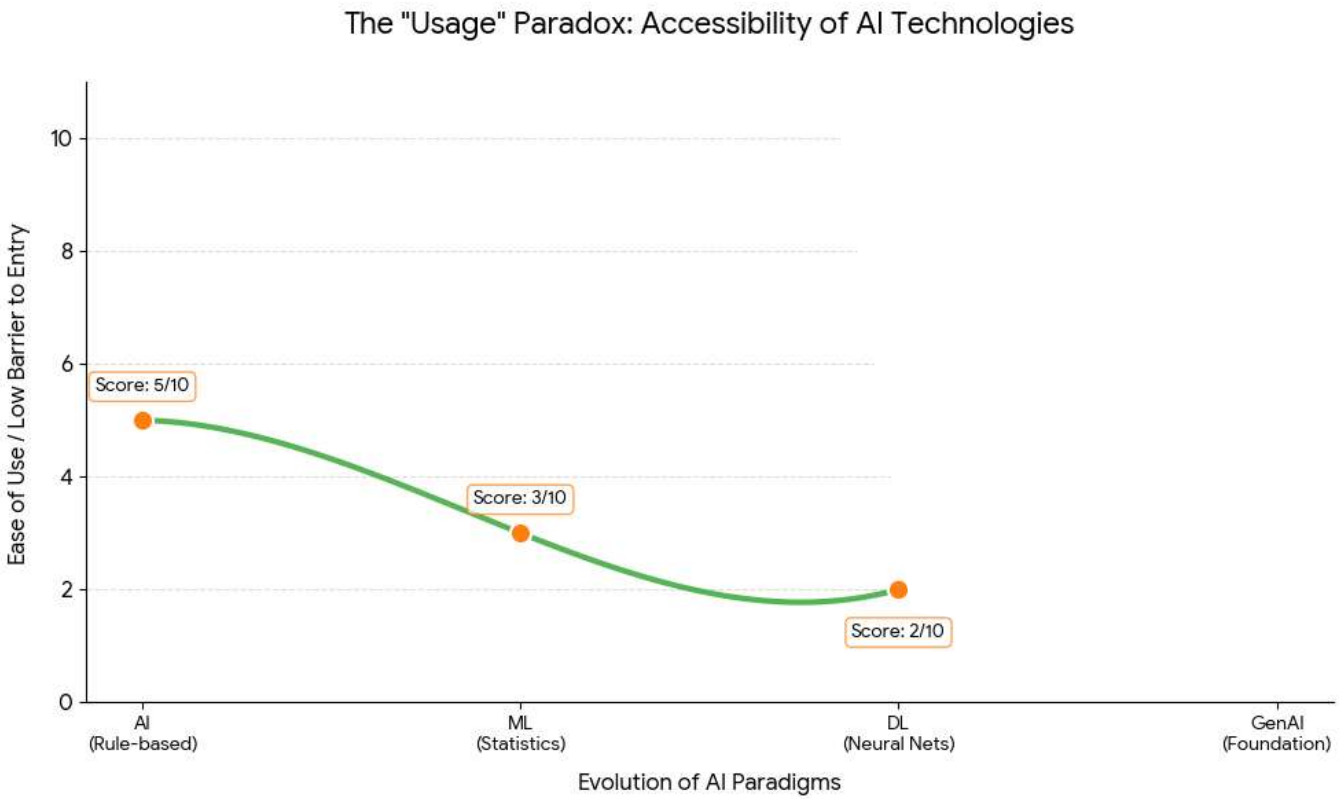
https://hbr.org/2025/06/recalculating-the-costs-and-benefits-of-gen-ai

GenAI Boom

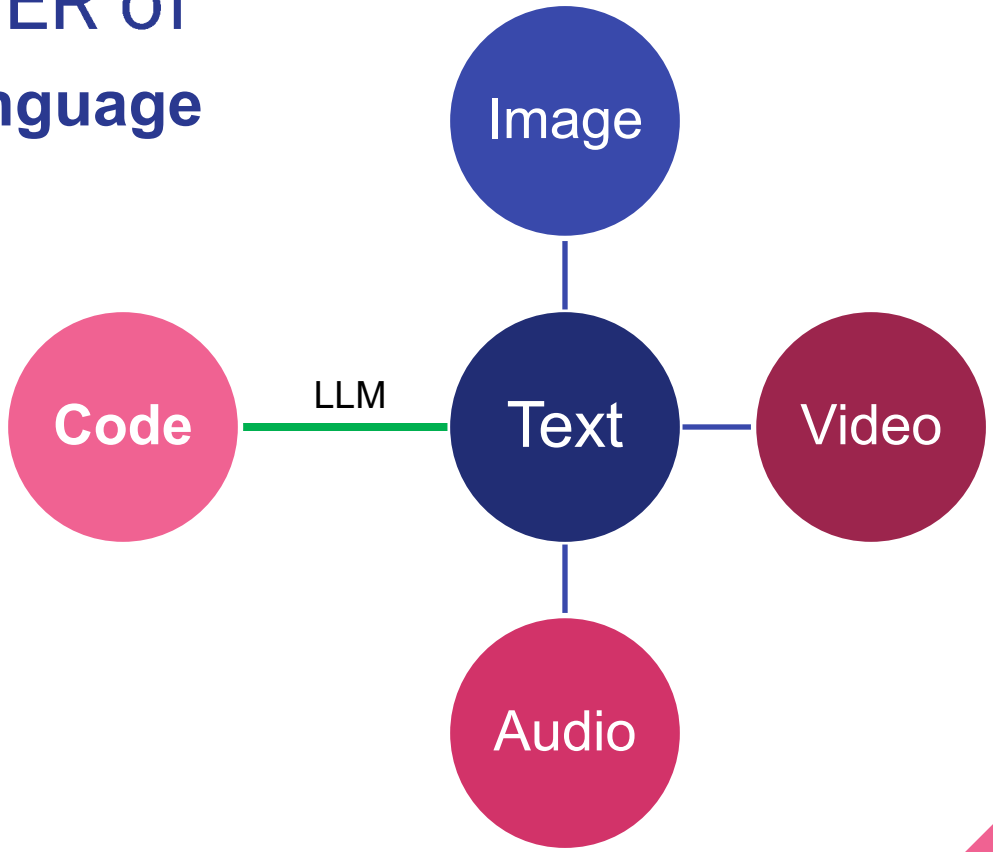

Evolution of AI
Artificial Intelligence
AI
Machine Learning
Deep Learning
Generative AI
Time

GenAI Boom




The "Usage" Paradox: Accessibility of AI Technologies
Ease of Use / Low Barrier to Entry
10
8
6
4
2
0
Score: 5/10
Score: 3/10
Score: 2/10
AI
(Rule-based)
ML
(Statistics)
DL
(Neural Nets)
GenAI
(Foundation)
Evolution of AI Paradigms

The POWER of
Natural Language



Gen AI model families

Acronym	Full name / type	Main modality	Typical use cases
LLM	Large Language Model	Text (sometimes code, structured text)	Chatbots, coding assistants, search, summarization, translation, agents.
SLM	Small Language Model	Text	On-device assistants, offline translation, domain bots, latency- and cost-sensitive workloads.
VLM	Vision–Language Model	Images + text (sometimes video)	Image captioning, visual QA, document/diagram understanding, grounding
LMM / MLLM	Multimodal (Large) Model	Multiple: text, images, sometimes audio/video	Chat over images, charts, UI; multimodal agents that “see” and “read”; video + text understanding.
LVM	Latent Variable Model (generative)	Depends on instance (images, text, etc.)	Representation learning, generation, anomaly detection, clustering; often used as building blocks.

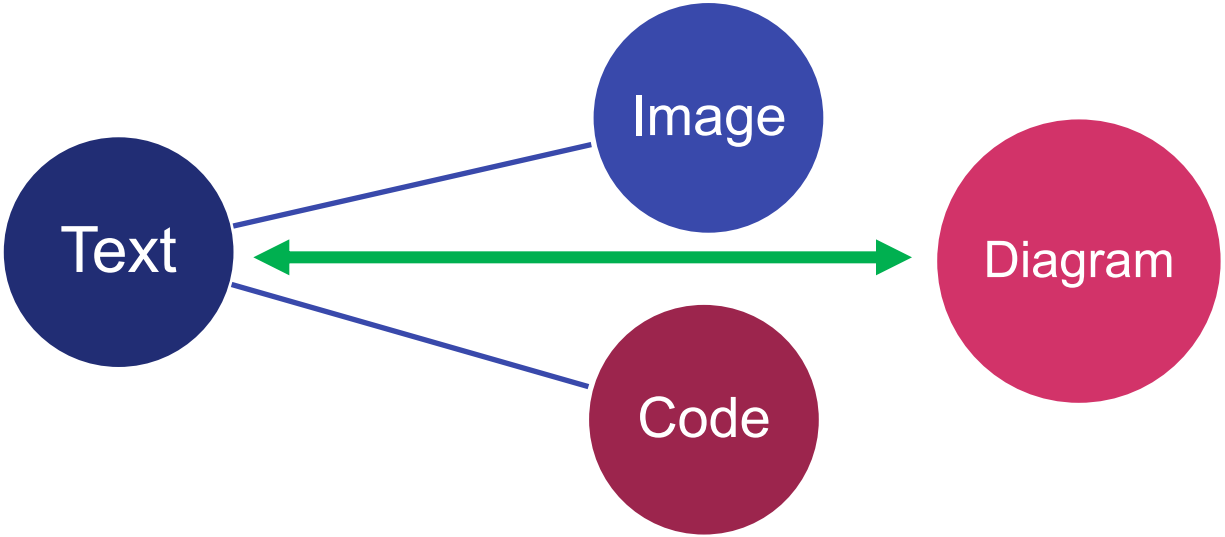
Gen AI model families

Acronym	Full name / type	Main modality	Typical use cases
GAN	Generative Adversarial Network	Mostly images, some audio/video	Image synthesis, style transfer, super-resolution, data augmentation.
Diffusion model	Denoising diffusion model	Primarily images, increasingly audio/video and 3D	Text-to-image (Stable Diffusion–style), image editing, video generation, high-fidelity synthesis.
MoE	Mixture of Experts	Usually text (can be multimodal)	Scaling large models cheaply, serving high-throughput language or multimodal workloads.
LAM	Large Action Model	Text + environment state / tools	Agents that operate UIs, call APIs, orchestrate workflows and tools.
RAG (pattern)	Retrieval-Augmented Generation	Text + external knowledge	Enterprise QA, chat over docs, up-to-date knowledge without full retraining.

Practical Use Cases

New possibilities

What is “Diagram as Text”?



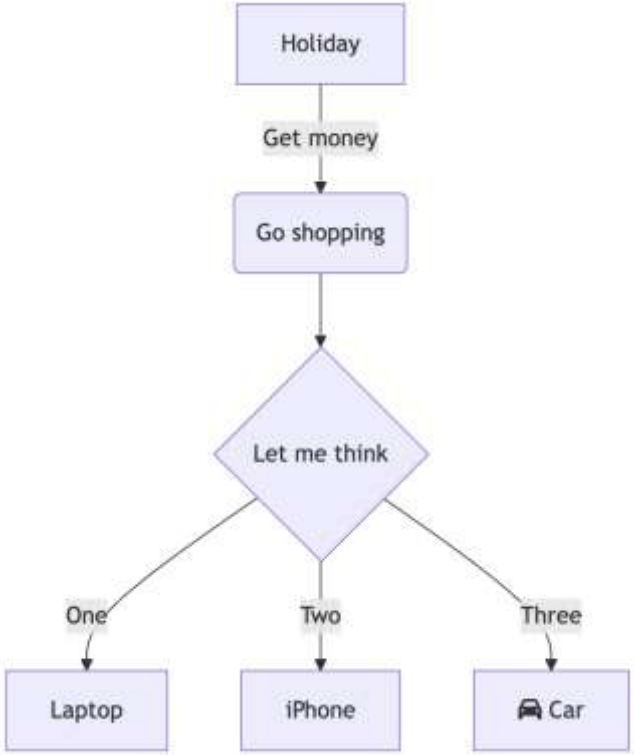
Text-to-Diagram/Text-to-chart is a ***text*** syntax that allows users to generate and render diagrams, charts, and visualizations from text.

Mermaid Diagrams

Mermaid Live Editor

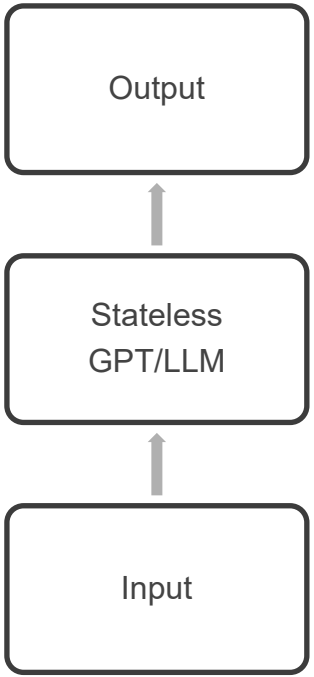
Code | Config | Docs

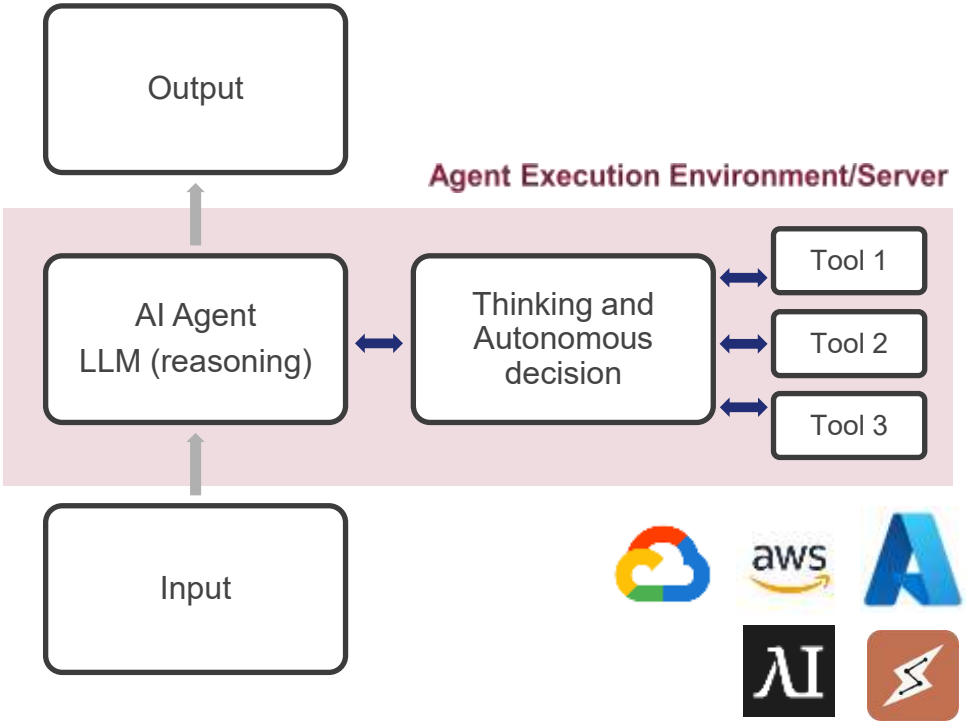
```
flowchart TD
    A[Holiday] -->|Get money| B(Go shopping)
    B --> C{Let me think}
    C -->|One| D[Laptop]
    C -->|Two| E[iPhone]
    C -->|Three| F[fa:fa-car Car]
```



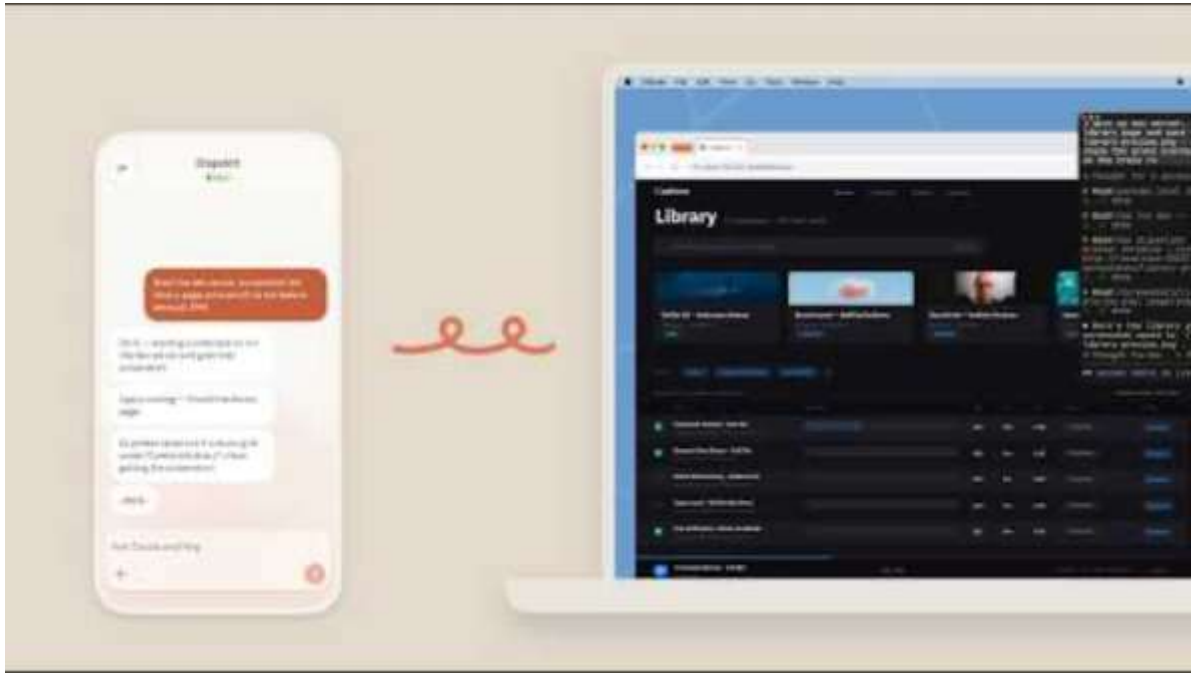
mermaid.live

What is an “Agent”? (Agentic AI)





Claude Cowork



🔗 **claude.com/cowork**

What is “AI Hallucination”?

NotebookLM Core Functionalities

Source Fundamentals

- Multi-format Source Support
- 100+ Languages
- Source Capacity
- Privacy & Security

Chat Capabilities

- Conversational Chat
- Cross-Referencing
- Citation and Grounding
- Pattern Recognition and Insights
- Customisable Conversational Style (Pro only)

Studio Features

- Audio Overview/Podcast
- Mind Maps
- Manual Note-Taking
- Other Resource Generation
- Saved Notes Panel
- Video Overview

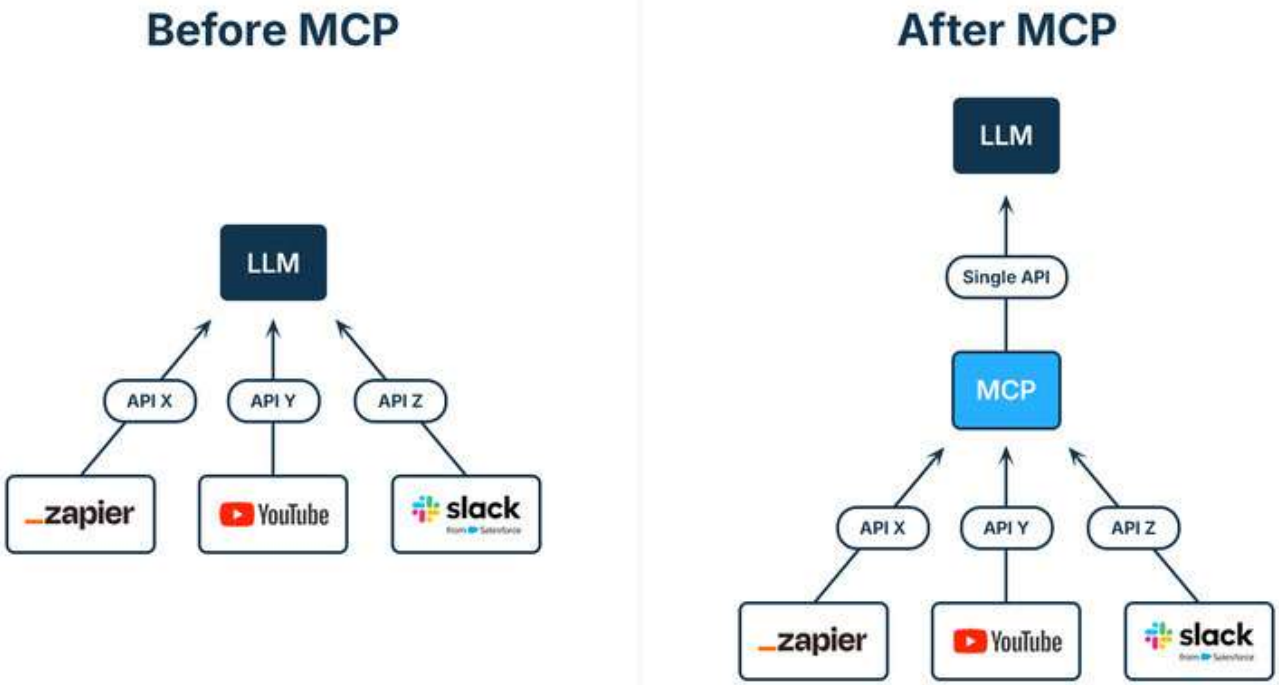
Collaboration Features

- Notebook Sharing
- Chat-Only Access
- User Analytics

notebooklm.google.com

devoteam

What is “MCP”?



https://www.civo.com/blog/what-is-mcp

Scite: AI for Research

scite_

Scite combines access to both Open Access and paywalled content with Smart Citations database that analyzes and classifies 1.6B+ citations across 280M+ sources.

Scite MCP: Enhance your AI tools with more accurate, grounded answers by connecting to Scite's literature database and Smart Citations.

scite.ai

Scite: AI for Research

scite_

scite_ Features Integrations Data Pricing Log In Start Free Trial

New Chat

RECENT

Start your free trial to save and access your chat history.

Start Free Trial

patented apparatus to study confined phase behavior

Fact checks (5 verified)

Title: Patented apparatus concepts for studying confined phase behavior of fluids

Executive summary

Unlock the complete response

Start free trial

Already have an account? Sign In

References (12) Search Strategy

Copy

Add references to dashboard

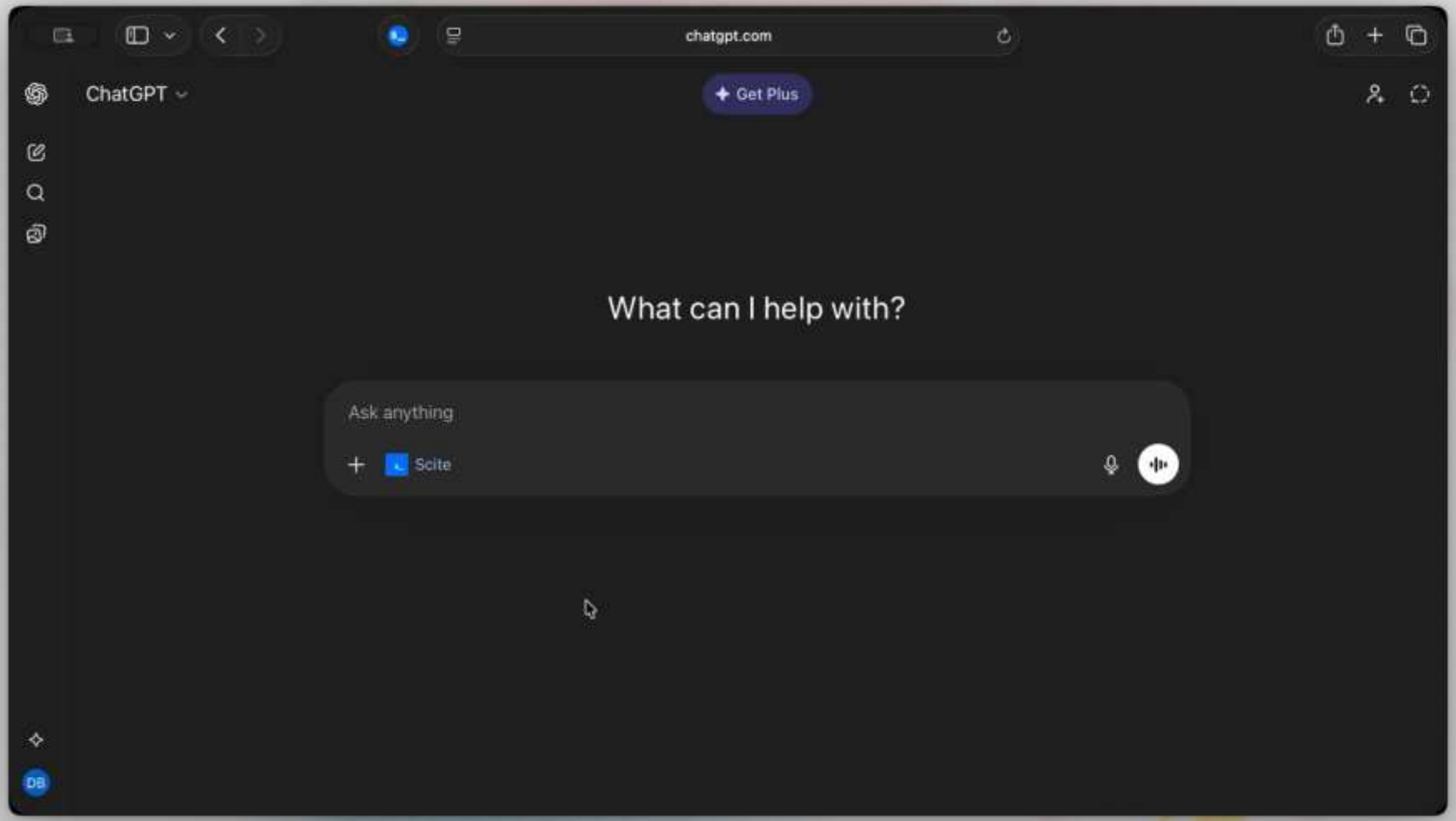
The effect of confinement on the phase behavior of propane in nanoporous media: an experimental study probing capillary condensation, evaporation, and hysteresis at varying pore sizes and temperatures

Rami M. Alloush[1], Keerti Vardhan Sharma[2], Mohammad Piri[3] 2024

Phys. Chem. Chem. Phys.

10 0 5 0

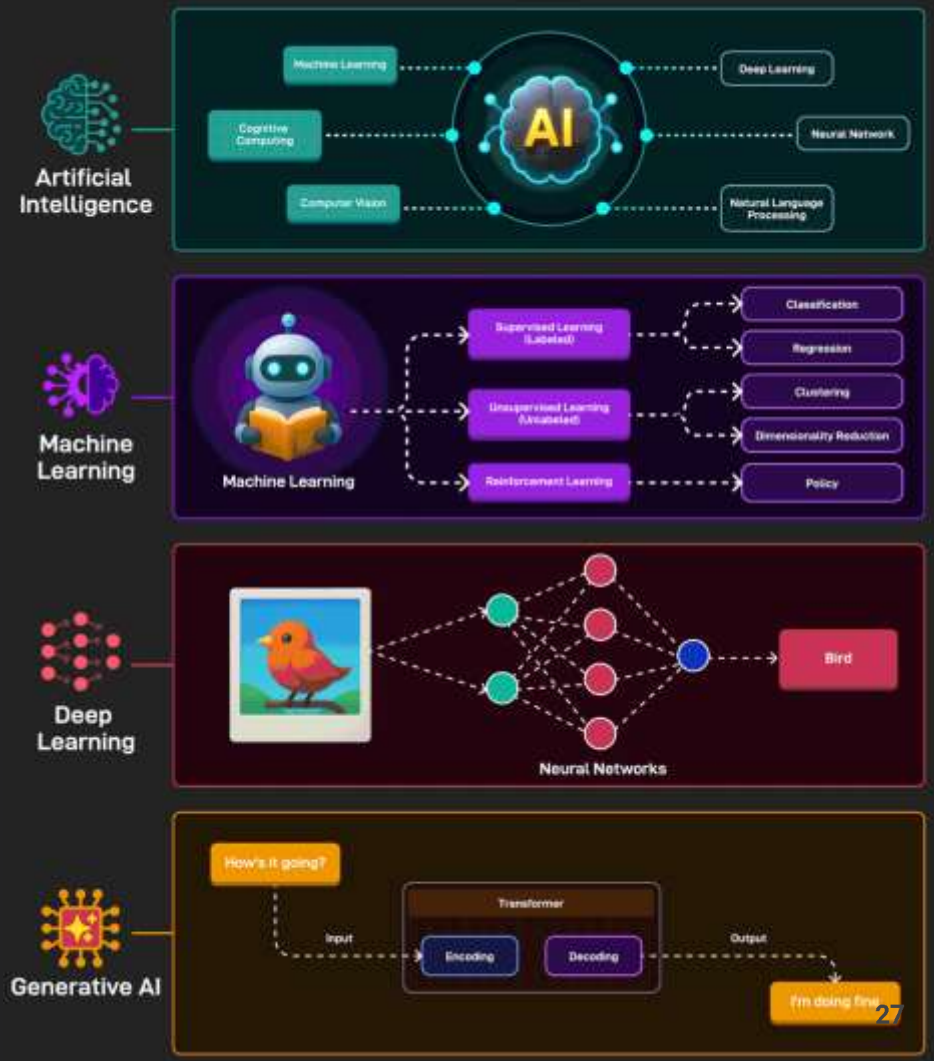
scite.ai


chatgpt.com
ChatGPT
Get Plus
What can I help with?
Ask anything
Scite
DB

Scenario

You feed 10 years of sensor data (wind speed, load, temperature) into an algorithm. Now, the system can predict the remaining useful life of a bridge with 95% accuracy.

- **The Answer: Machine Learning (ML)**
- **Why?** The algorithm learns and finds patterns in structured data.

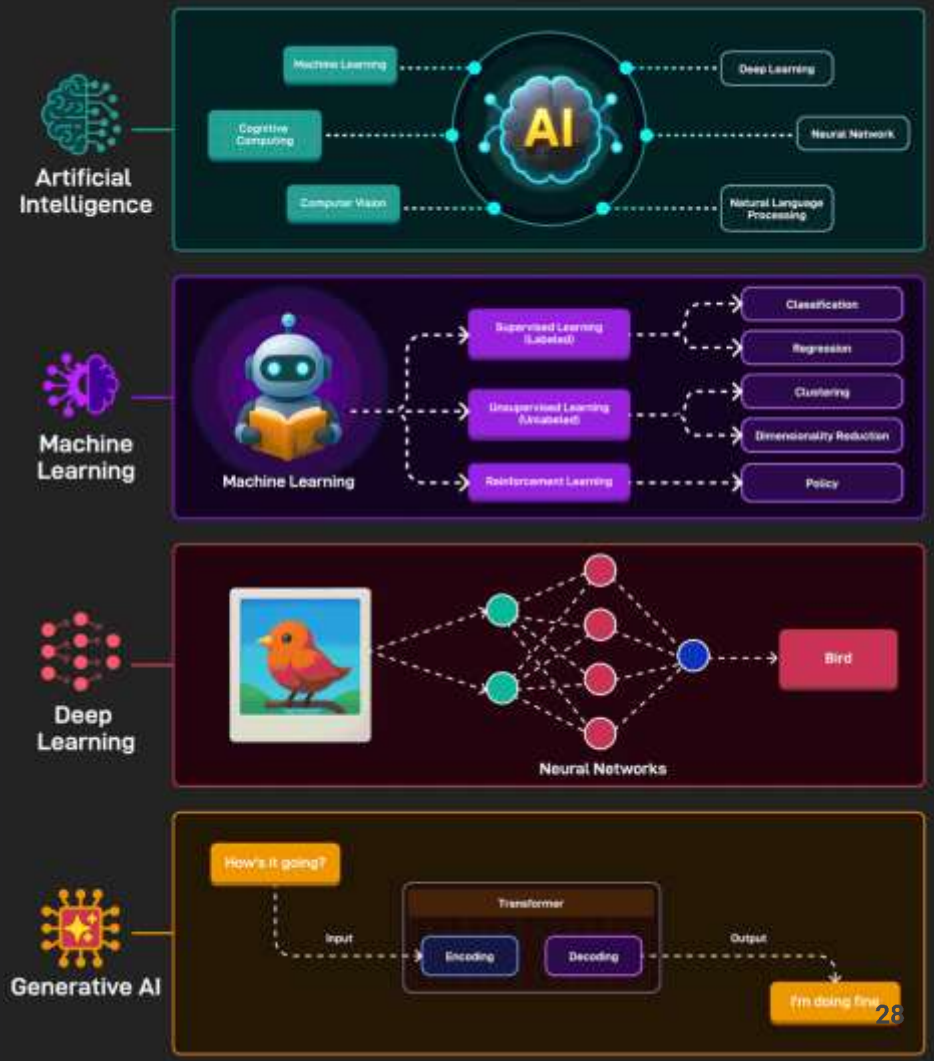


https://blog.bytebytego.com/p/ep168-ai-vs-machine-learning-vs-deep

Scenario

You provide load constraints; the AI creates 50 new, organic bracket designs that minimize weight.

- **The Answer: Generative AI (GenAI)**
- **Why?** It isn't just classifying a design as "Good" or "Bad"; it is synthesizing brand new geometry.



https://blog.bytebytego.com/p/ep168-ai-vs-machine-learning-vs-deep

Thank You :)

www.RamiAlloush.com